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Old and New Methods for Lost-Sales Inventory Systems

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We consider the notoriously difficult discrete-time inventory model with stochastic demands, a constant lead time, and lost sales. We show that the effective state space is a relatively manageable compact set. Then, we test various plausible heuristics. We find that several perform reasonably well, although none is perfect. However, the standard base-stock policy (a direct analogue of the optimal policy for a backlog system) performs badly. We also show that the optimal cost is increasing in the lead time.

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1. Introduction

We have a fairly good understanding of inventory systems where stockouts are backlogged. We know good policies for these systems, and we understand their behavior. For example, for a basic system with stationary data, constant lead time, linear order cost, and independent random demands, we know that the optimal policy has a simple form, a base-stock policy, and we can readily compute the optimal one. Also, we know that the optimal policy and the overall cost depend on the physical system's parameters roughly through the standard deviation of lead-time demand—the one-period demand's standard deviation times the square root of the lead time. Finally, we know how to extend these results in several directions—a fixed as well as a linear order cost, stochastic lead times, correlated and/or nonstationary demands, etc. (see, e.g., Porteus 2002 and Zipkin 2000).

Many real systems have lost sales, or equivalently expediting, not backorders (see, e.g., Gruen et al. 2002). Our understanding of such systems is much weaker. The problem was formulated a half century ago by Karlin and Scarf (1958). They and Morton (1969) established some basic structural results. Still, it is much harder to compute an optimal policy, even for the simplest systems (with stationary data, etc.). The state space grows rapidly as the lead time increases. Partly for this reason, plausible heuristics have been tested only on a limited range of systems. Moreover, we know almost nothing about how good policies and their costs behave with respect to the system's parameters. Extensions to more complex systems are few (see, e.g., Nahmias 1979, Cohen et al. 1988, Johansen and Thorstenson 1993, and Melchioris et al. 2000).

We do know that a lost-sales system is fundamentally different from one with backorders. For example, provided the mean demand is finite, the optimal cost in the lost-sales system is bounded as a function of the lead time and

the demand variance. (To see this, just consider the policy of never ordering. Its average cost is the lost-sales penalty cost times the mean demand.) The backorders model, in contrast, has unbounded cost—that is one implication of the square-root relation mentioned above.

We show that, under any plausible policy, the effective state space is a relatively manageable compact set. It still grows rapidly in the lead time, but less so than it would otherwise. Armed with this technique, plus modern computer technology, we are able to solve models with longer lead times than any reported previously. We then evaluate various plausible heuristics. Some of these were proposed decades ago, but others quite recently. We find that several perform reasonably well, although none is perfect. However, the standard base-stock policy (a direct analogue of the optimal policy for a backlog system) performs poorly.

We also show that the optimal cost is increasing in the lead time. This fact is not surprising, but it is useful to know. (This monotonicity result applies to any system with lagged control, not just the lost-sales system.)

This paper is organized as follows. Section 2 presents the formulation. Section 3 describes the heuristics. Section 4 derives the state-space result. Section 5 presents the lead-time monotonicity result. Section 6 reports numerical results.

2. Formulation

Consider the standard, single-item inventory system in discrete time with lost sales. The demands and state variables can be either integers or continuous. Denote

L = order lead time, $L > 0$.

t = time index, $t = 1, \dots, T$.

d_t = demand in period t .

w_t = inventory at time t , before the order due at t arrives.

z_t = order at time t .

y_t = inventory at time t , after the order due at t arrives
= $w_t + z_{t-L}$.

$u_t = y_t - d_t$.

$x_{0t} = y_t$.

$x_{1t} = z_{t+1-L}$

...

$x_{L-1,t} = z_{t-1}$.

$\mathbf{x}_t = (x_{0t}, x_{1t}, \dots, x_{L-1,t})$.

Assume that the d_t are independent and nonnegative. The state of the system is the L -vector $\mathbf{x}_t$. The dynamics are

$$\begin{aligned} x_{0,t+1} &= y_{t+1} = w_{t+1} + z_{t+1-L} \\ &= [y_t - d_t]^+ + z_{t+1-L} \\ &= [x_{0t} - d_t]^+ + x_{1t}, \end{aligned}$$

$$\begin{aligned} x_{1,t+1} &= z_{t+2-L} = x_{2t} \\ &\dots \end{aligned}$$

$$x_{L-2,t+1} = z_{t-1} = x_{L-1,t},$$

$$x_{L-1,t+1} = z_t,$$

or

$$\mathbf{x}_{t+1} = ([x_{0t} - d_t]^+ + x_{1t}, x_{2t}, \dots, x_{L-1,t}, z_t).$$

For simplicity, assume that the data (cost factors and demand distributions) are stationary, and the order cost, holding cost, and stockout-penalty cost are linear. Let

c = unit cost of procurement,

$\hat{h}$ = unit cost of holding inventory,

p = unit cost penalty of lost sales, and

γ = discount rate.

Let d denote a generic demand, $\mathbf{x}$ a generic state vector, and z a generic order quantity. Given this period's $\mathbf{x}$, z , and d , let $\mathbf{x}_+$ denote the next period's state, that is,

$$\mathbf{x}_+ = ([x_0 - d]^+ + x_1, x_2, \dots, x_{L-1}, z).$$

Similarly, when t is understood, use the subscript $+l$ to indicate l periods later; for example, $u_{+L} = u_{t+L}$.

Given $u_t = u$, the holding-penalty cost at the end of period t is

$$\hat{q}(u) = \hat{h}u^+ + pu^-.$$

The expected cost, viewed from the beginning of the period, given $y_t = y$, is

$$\hat{q}^0(y) = E[\hat{q}(y - d)].$$

Define

$\hat{f}_t(\mathbf{x})$ = optimal cost from time t onwards,
starting in state $\mathbf{x}_t = \mathbf{x}$.

These functions satisfy the recursion

$$\hat{f}_t(\mathbf{x}) = \min_{z \geq 0} \{ \gamma^L c z + \hat{q}^0(y) + \gamma E[\hat{f}_{t+1}(\mathbf{x}_+)] \}.$$

For a finite time horizon T , the following terminal condition is commonly assumed:

$$\hat{f}_{T+1}(\mathbf{x}) = 0. \quad (1)$$

Now perform some standard transformations (see Zipkin 2000, Chapter 9): First, assume that the process continues and holding-penalty costs accumulate through period $T+L$. This means a terminal condition different from (1). Next, for each t , ignore the holding-penalty costs through period $t+L-1$ (which do not depend on the current order z), but assign to period t the holding-penalty cost in period $t+L$. Also, charge the system for the procurement of any inventory at the beginning of period $t+L$ (that is, add the cost $\gamma^L c E[w_{+L}]$, which depends on $\mathbf{x}$ but not z), but suppose that any remaining inventory at time $T+L+1$ can be sold at cost c . Let $f_t(\mathbf{x})$ denote the optimal cost under this new accounting scheme. Some algebra reveals that these functions satisfy the recursion

$$f_t(\mathbf{x}) = \min_{z \geq 0} \{ q(\mathbf{x}, z) + \gamma E[f_{t+1}(\mathbf{x}_+)] \}, \quad (2)$$

$$f_{T+1}(\mathbf{x}) = 0,$$

where

$$q(\mathbf{x}, z) = \gamma^L \{ c E[y_{+L}] + h E[u_{+L}^+] + p E[u_{+L}^-] \}$$

with $h = \hat{h} - \gamma c$.

For $T = \infty$, provided $\gamma < 1$, the optimal cost uniquely satisfies the functional equation

$$f(\mathbf{x}) = \min_{z \geq 0} \{ q(\mathbf{x}, z) + \gamma E[f(\mathbf{x}_+)] \}. \quad (3)$$

Let $f^*(\mathbf{x})$ denote the solution and $z^*(\mathbf{x})$ an optimal policy. We shall work mostly with this model. (Janakiraman and Muckstadt 2004 show that this model is equivalent to one with $c = 0$. We have retained c , however, to keep its role transparent.)

The one-period cost q can be computed in a recursive manner: Let $\mathbf{x}^l = (x_0, \dots, x_l)$ denote the subvector consisting of the first $l+1$ components of $\mathbf{x}$. Therefore, $\mathbf{x}^{L-1} = \mathbf{x}$. Let us also write $z = x_L$ and $\mathbf{x}^L = (\mathbf{x}, z)$. Letting

$$q^0(y) = cy + hE[(y - d)^+] + pE[(y - d)^-]$$

and

$$q^l(\mathbf{x}^l) = \gamma^l E[q^0(y_{+l})], \quad l = 1, \dots, L,$$

we have

$$q^l(\mathbf{x}^l) = \gamma^l E[q^{l-1}(\mathbf{x}_+^{l-1})], \quad l = 1, \dots, L,$$

and $q(\mathbf{x}, z) = q^L(\mathbf{x}^L)$.

3. Heuristics

We now briefly describe the main heuristics that have been proposed for this problem.

The *myopic policy* (Morton 1971) selects z to minimize $q(\mathbf{x}, z)$, the one-period cost above. In other words, it solves the problem with horizon $T = 1$. In the case of integer demands and state variables,

$$\frac{1}{\gamma^L} \Delta_z q(\mathbf{x}, z) = (c + h) - (p + h) \Pr\{u_{+L} < 0\}.$$

The policy chooses the smallest $z \geq 0$ such that $\Delta_z q(\mathbf{x}, z) \geq 0$, that is,

$$\Pr\{u_{+L} < 0\} \leq \frac{c + h}{p + h} \doteq \bar{\theta}.$$

For continuous demands and state variables,

$$\frac{1}{\gamma^L} \frac{\partial}{\partial z} q(\mathbf{x}, z) = (c + h) - (p + h) \Pr\{u_{+L} < 0\}.$$

The myopic policy sets $z = 0$ if $(\partial/\partial z)q(\mathbf{x}, 0) \geq 0$, and otherwise solves $(\partial/\partial z)q(\mathbf{x}, z) = 0$, that is,

$$\Pr\{u_{+L} < 0\} = \bar{\theta}.$$

Several investigators (e.g., Bollapragada and Morton 1999 and Iida and Zipkin 2006) studying different but related models have found that when the myopic policy is not optimal, it can be improved considerably by extending the horizon from $T = 1$ to a few more periods. Then, use the policy that is optimal for $t = 1$ as a stationary policy for the infinite-horizon problem. Call this the *myopic- T policy* (so myopic-1 = myopic).

Morton (Equation (83) on page 594 of Morton 1969) derives bounds on the optimal policy and uses them to construct a simpler alternative heuristic. (That analysis covers the discounted-cost case. The same results hold for average costs; see Huh and Janakiraman 2005.) Define $v_l = \sum_{m=l}^{L-1} x_m$, $l = 0, \dots, L-1$, $v_L = 0$, and $d_{[l, L]} = \sum_{m=l}^L d_m$. Thus, each v_l is a partial sum of the order pipeline variables, starting with the most recent order x_{L-1} and then adding in successively older ones. For integer demand, set

$$\bar{s}_l = \min\{s: \Pr\{d_{[l, L]} > s\} \leq \bar{\theta}\}, \quad l = 0, \dots, L.$$

For continuous demand, this simplifies to $\Pr\{d_{[l, L]} > \bar{s}_l\} = \bar{\theta}$. Thus, $\bar{s}_l$ is the $(1 - \bar{\theta})$ fractile of the distribution of $(L - l + 1)$ -period demand. Morton shows that for $\mathbf{x}$ with $z^*(\mathbf{x}) > 0$,

$$v_l + z^*(\mathbf{x}) \leq \bar{s}_l, \quad l = 0, \dots, L. \quad (4)$$

This condition imposes several upper bounds on an optimal order, one for each l , for every state. The heuristic policy sets z to the largest value consistent with these bounds, that is,

$$z(\mathbf{x}) = [\min\{\bar{s}_l - v_l, l = 0, \dots, L\}]^+. \quad (5)$$

Let us call this the *standard vector base-stock policy*. It can be viewed as a vector analogue to a base-stock policy. Set $\mathbf{v} = (v_l)_{l=0}^L$, $\bar{\mathbf{s}} = (\bar{s}_l)_{l=0}^L$, and let $\mathbf{e}$ denote the L -vector of 1s. The policy compares the current vector $\mathbf{v}$ to the fixed policy vector $\bar{\mathbf{s}}$. It orders only when $\mathbf{v}$ is less than $\bar{\mathbf{s}}$ in every component, and then it orders the largest z , such that the whole vector $\mathbf{v} + z\mathbf{e}$ does not exceed $\bar{\mathbf{s}}$ —that is, such that $\mathbf{v} + z\mathbf{e} \leq \bar{\mathbf{s}}$.

Let us further define the class of policies of this form as *vector base-stock policies*, using any vector $\mathbf{s}$ in place of $\bar{\mathbf{s}}$ in (5). (We call the one with $\mathbf{s} = \bar{\mathbf{s}}$ standard because this is the one Morton originally proposed. Also, its parameters $\bar{\mathbf{s}}$ are obtained from the original data in a simple way.) There are many such policies. How might one find a good one? Here is one idea (following a hint in Morton 1969): Use the procedure above for $\bar{\mathbf{s}}$, but with other values θ in place of $\bar{\theta}$. This reduces the problem to a one-dimensional search over θ .

An ordinary *base-stock policy* is a special case, having all s_l equal. The *standard base-stock policy* uses all $s_l = \bar{s}_0$. This policy is optimal for the backlog system. Many texts present it as a reasonable heuristic for the lost-sales system.

Levi et al. (2008) propose a heuristic called the *dual-balancing policy*. They show that its cost is at most twice the optimal cost. (This is the first demonstration of a worst-case error bound for the lost-sales system.) The calculations involved are similar to those of the myopic policy, but they also estimate holding costs in future periods, not just the current one, and the results are used differently. Instead of minimizing the sum of all expected costs, this method aims to equate (“balance”) the purchase-plus-holding costs and penalty costs.

Reiman (2004) has recently studied the stationary system in continuous time. He analyzes an astonishingly simple and seemingly dull-witted policy—order at fixed intervals of time, no matter what happens. He shows that this policy is sometimes better than the best base-stock policy. In fact, this is always so for a sufficiently long lead time. The natural discrete-time analogue is a *constant-order policy*—order the same amount every period. Provided this amount is less than the mean demand, the system under this policy is stable, with a finite cost. This cost is easy to compute. One can then optimize over the order size. The cost of this policy, like that of never ordering, does not depend on the lead time. It thus provides a sharper upper bound on the optimal cost. (It does depend on the demand distribution, however.) Reiman does not intend this policy as a serious candidate for practical use. Rather, it is an analytical device for probing the behavior of the system, and the bound provides a useful benchmark.

4. State Reduction

Morton (1969) demonstrates that under an optimal policy, the state remains in a bounded set. We now apply a similar argument to show that the same is true for any plausible

policy, including the heuristics above. This fact permits a substantial reduction of the state space.

For any $(L + 1)$ -vector $\mathbf{s}$ with $s_0 \geq s_1 \geq \dots \geq s_L \geq 0$, define

$$\mathbf{X}(\mathbf{s}) = \{\mathbf{x} \geq \mathbf{0}: v_l \leq s_l, l = 0, \dots, L - 1\}.$$

Let $\mathbf{Z}(\mathbf{s})$ denote the set of policies such that $z(\mathbf{x}) = 0$ for $\mathbf{x} \notin \mathbf{X}(\mathbf{s})$, and

$$v_l + z(\mathbf{x}) \leq s_l, \quad l = 0, \dots, L, \mathbf{x} \in \mathbf{X}(\mathbf{s}).$$

Note that $\mathbf{X}(\mathbf{s})$ and $\mathbf{Z}(\mathbf{s})$ are increasing in $\mathbf{s}$, that is, $\mathbf{s} \leq \mathbf{s}'$ implies $\mathbf{X}(\mathbf{s}) \subseteq \mathbf{X}(\mathbf{s}')$ and $\mathbf{Z}(\mathbf{s}) \subseteq \mathbf{Z}(\mathbf{s}')$.

The bounds (4) indicate that any optimal policy is in $\mathbf{Z}(\bar{\mathbf{s}})$, and so is the standard vector base-stock policy, by construction. Morton (1971) observes that the myopic policy is too. Any plausible policy sets $z(\mathbf{x}) = 0$ for $\mathbf{x}$ sufficiently large, and therefore the policy is in $\mathbf{Z}(\mathbf{s})$ for $\mathbf{s}$ sufficiently large.

CLAIM 1. For any policy in $\mathbf{Z}(\mathbf{s})$, all states outside $\mathbf{X}(\mathbf{s})$ are transient.

PROOF. If the process starts outside $\mathbf{X}(\mathbf{s})$, it must eventually reach $\mathbf{X}(\mathbf{s})$ because $z(\mathbf{x}) = 0$ until then. Therefore, suppose that $\mathbf{x} \in \mathbf{X}(\mathbf{s})$. Then,

$$(v_{L-1})_+ = z(\mathbf{x}) \leq s_L \leq s_{L-1},$$

$$(v_{l-1})_+ = v_l + z(\mathbf{x}) \leq s_l \leq s_{l-1}, \quad l = 2, \dots, L - 1,$$

$$(v_0)_+ \leq v_0 + z(\mathbf{x}) \leq s_0,$$

so $\mathbf{x}_+ \in \mathbf{X}(\mathbf{s})$. Once the process enters $\mathbf{X}(\mathbf{s})$, it never leaves. $\square$

Consequently, to solve the dynamic program, one can restrict attention to the compact state space $\mathbf{X}(\bar{\mathbf{s}})$ and policies $\mathbf{Z}(\bar{\mathbf{s}})$. In fact, the effective state space is even smaller. Because $z(\mathbf{x}) \leq \bar{s}_L$, so are all of x_l , $l = 1, \dots, L - 1$. Also, because $z(\mathbf{x}) + x_{L-1} \leq \bar{s}_{L-1}$, we also have $x_{L-1} + x_{L-2} \leq \bar{s}_{L-1}, \dots, x_2 + x_1 \leq \bar{s}_{L-1}$, etc.

5. Monotonicity

Next, we argue that the system cost is nondecreasing in L . Consider two systems, identical in every way except that one has lead time L and the other has lead time $L - 1$. Call these systems L and $L - 1$, for short. Suppose that we know an optimal policy for system L . For system $L - 1$, in each period, based on the information available, fix the order one will place in the next period, and set this quantity equal to system L 's current optimal order. If the two systems face the same demand sequence, this policy produces precisely the same events in system $L - 1$ as the original policy in system L . Thus, these policies have the same costs, hence the same expected costs, in the two systems. An optimal policy for system $L - 1$ must cost the same or less.

Note that the policy constructed for system $L - 1$ is not Markovian; the order is not a function just of the state $\mathbf{x}^{L-2}$. To implement it requires expanding the state to $\mathbf{x}^{L-1}$ to remember the right order in the next period. Still, it is a bona fide policy—it determines each order as a function of the history of the process.

Let us explore the implications in more detail. First, consider the discounted-cost case. System L begins to accumulate costs only L periods in the future, but system $L - 1$ also incurs cost one period earlier. The cost in that period is $q^{L-1}(\mathbf{x}^{L-1})$. Letting $f^{*L}(\mathbf{x}^{L-1})$ denote the optimal cost for system L , we have

$$f^{*L-1}(\mathbf{x}^{L-2}) \leq q^{L-1}(\mathbf{x}^{L-1}) + f^{*L}(\mathbf{x}^{L-1}).$$

Turning this around, we obtain a lower bound on f^{*L} :

$$f^{*L}(\mathbf{x}^{L-1}) \geq f^{*L-1}(\mathbf{x}^{L-2}) - q^{L-1}(\mathbf{x}^{L-1}).$$

Substitute this into the optimality equation (3):

$$f^{*L}(\mathbf{x}^{L-1}) \geq \min_{x_L \geq 0} \{q^L(\mathbf{x}^L) + \gamma E[f^{*L-1}(\mathbf{x}_+^{L-2}) - q^{L-1}(\mathbf{x}_+^{L-1})]\}.$$

However, $q^L(\mathbf{x}^L) = \gamma E[q^{L-1}(\mathbf{x}_+^{L-1})]$, so these two terms cancel. The remaining term $E[f^{*L-1}(\mathbf{x}_+^{L-2})]$ is independent of x_L . We are left with

$$f^{*L}(\mathbf{x}^{L-1}) \geq \gamma E[f^{*L-1}(\mathbf{x}_+^{L-2})]. \quad (6)$$

In this way, we can use the solution of system $L - 1$ to bound the cost of system L .

In the average-cost case, the argument above proves that the average cost is nondecreasing in L . That is, given an average-cost optimal policy for system L , one can construct a feasible policy for system $L - 1$ with the same average cost. Therefore, the minimal average cost for system $L - 1$ can only be less.

This suggests a numerical strategy. First, solve system $L - 1$. Then, use the best of the lower bounds as a starting solution for system L . One could even start with $L = 1$ and solve systems with progressively larger L .

Clearly, this argument applies to any system with control lag, not just this one.

6. Numerical Results

We solved a small set of test problems with lead times up to four. (This is twice the previous record, set by Morton 1971.) All have $c = 0$ and $\hat{h} = 1$, so $h = 1$ and $\gamma = 0.995$. (Recall that any instance of the model is equivalent to one with $c = 0$.) We tried four values of p , 4, 9, 19, and 39, corresponding to $\bar{\theta} = 0.2, 0.1, 0.05$, and 0.025 . We used two demand distributions, Poisson and geometric, both with mean five.

The sizes of the effective state spaces, using the state-reduction technique, are given in the first row of each part of Tables 1 and 2. Evidently, they grow rapidly in L . It

Table 1. Problem size and effort: Poisson demand.

	Lead time			
	1	2	3	4
(a) $p = 4$				
States	14	124	1,152	9,984
Standard				
Iterations	11	13	18	36
Seconds			3	47
Bound				
Iterations	11	13	18	31
Seconds			3	41
(b) $p = 9$				
States	15	153	1,539	15,309
Standard				
Iterations	9	14	18	25
Seconds		1	4	52
Bound				
Iterations	9	13	17	25
Seconds		1	3	52
(c) $p = 19$				
States	16	185	2,000	20,500
Standard				
Iterations	7	12	17	21
Seconds			5	57
Bound				
Iterations	7	12	16	20
Seconds			4	57
(d) $p = 39$				
States	18	209	2,420	27,945
Standard				
Iterations	6	11	14	20
Seconds			5	76
Bound				
Iterations	6	11	14	19
Seconds			5	74

would be much harder to solve these models without such state reductions (Puterman 1994, §6.10), and we did not try.

We solved the models using the standard value-iteration method (Puterman 1994, §6.3) with no enhancements such as splitting or action elimination. We did use the fact that, at each iteration and for each state, the function to be minimized is convex in z . (This is implied by Morton's 1969 analysis.) To find its minimum, therefore, we used a simple local search. We used the bounds on f^* of Porteus (1971), which refine those of MacQueen (1966), to test for termination. The program was written in GAMS and run on a standard laptop computer.

Tables 1 and 2 give the number of iterations and time required to solve each case. We tried two initial solutions, 0 (the standard approach) and the lower bound (6). Evidently, (6) helps only a little; it helps more in the cases with geometric demand.

The cases with geometric demand are harder than those with Poisson demand. This makes sense. The tail of the geometric distribution decays more slowly, so the state bounds $\bar{s}_l$ are larger. Also, a larger p makes a problem harder, again because the $\bar{s}_l$ are larger.

Table 2. Problem size and effort: Geometric demand.

	Lead time			
	1	2	3	4
(a) $p = 4$				
States	16	171	1,701	16,767
Standard				
Iterations	11	16	23	23
Seconds			5	176
Bound				
Iterations	11	16	22	22
Seconds			4	167
(b) $p = 9$				
States	21	299	4,056	52,722
Standard				
Iterations	10	16	22	28
Seconds		1	12	197
Bound				
Iterations	10	15	20	25
Seconds		1	12	183
(c) $p = 19$				
States	26	442	7,514	122,363
Standard				
Iterations	10	17	22	27
Seconds		1	26	513
Bound				
Iterations	10	16	20	25
Seconds		1	23	468
(d) $p = 39$				
States	31	609	11,884	228,581
Standard				
Iterations	9	15	21	25
Seconds	1	2	42	1,004
Bound				
Iterations	9	14	19	23
Seconds	1	1	37	894

To compare policies, we switch to the average-cost criterion. It is easier to interpret—it is just a number, whereas the discounted cost is a function. We resolved each instance with the new objective. Tables 3 and 4 show the optimal average cost for each case. They also present the costs and percentages over the optimal of various heuristics. The bottom row of each part shows the cost of the best constant-order policy (which, remember, is independent of L).

To find the best base-stock policy, we used a simple local-search technique: At each step, change the base-stock level by ± 1 , and stop when no further improvement is possible. Because the average cost is convex as a function of the base-stock level (Downs et al. 2001), this method suffices. (No doubt there are faster methods.) We used a similar technique to find the best constant-order policy, even though we know of no theoretical guarantee that the resulting policy is truly the best. To select a vector base-stock policy, we performed a local search over θ , again with no assurance of optimality. We call the result a *better* vector base-stock policy.

Base-stock policies do not perform well. The standard policy's cost is high, and it grows considerably in the lead

Table 3. Average cost: Poisson demand.

Policy	Lead time							
	1		2		3		4	
(a) $p = 4$								
Optimal	4.04		4.40		4.60		4.73	
Myopic	4.11	1.7%	4.56	3.7%	4.84	5.3%	5.06	7.1%
Myopic-2	4.04	0.0%	4.41	0.2%	4.64	0.8%	4.82	1.9%
Dual-balancing	4.15	2.7%	4.41	0.4%	4.62	0.4%	4.73	0.1%
Std. vector base-stock	4.31	6.6%	4.80	9.3%	5.51	19.9%	5.83	23.4%
Better vector base-stock	4.07	0.6%	4.45	1.2%	4.63	0.7%	4.80	1.5%
Std. base-stock	4.39	8.6%	4.94	12.5%	5.77	25.5%	6.14	29.8%
Best base-stock	4.16	3.0%	4.64	5.5%	4.98	8.2%	5.20	9.9%
Best constant				5.27				
(b) $p = 9$								
Optimal	5.44		6.09		6.53		6.84	
Myopic	5.45	0.2%	6.22	2.1%	6.80	4.1%	7.20	5.3%
Myopic-2	5.44	0.0%	6.10	0.2%	6.57	0.6%	6.92	1.2%
Dual-balancing	5.73	5.4%	6.38	4.7%	6.78	3.8%	6.99	2.3%
Std. vector base-stock	5.51	1.4%	6.45	5.9%	7.31	12.0%	8.16	19.3%
Better vector base-stock	5.48	0.8%	6.17	1.3%	6.64	1.7%	6.95	1.7%
Std. base-stock	5.56	2.2%	6.55	7.6%	7.47	14.4%	8.37	22.5%
Best base-stock	5.55	2.0%	6.32	3.7%	6.86	5.1%	7.27	6.4%
Best constant				10.27				
(c) $p = 19$								
Optimal	6.68		7.66		8.36		8.89	
Myopic	6.69	0.2%	7.77	1.4%	8.56	2.4%	9.18	3.3%
Myopic-2	6.68	0.0%	7.67	0.1%	8.40	0.4%	8.96	0.8%
Dual-balancing	7.48	12.1%	8.45	10.2%	9.19	9.9%	9.65	8.6%
Std. vector base-stock	6.71	0.5%	8.08	5.4%	9.09	8.7%	9.57	7.7%
Better vector base-stock	6.69	0.2%	7.76	1.3%	8.47	1.3%	8.99	1.2%
Std. base-stock	6.73	0.8%	8.15	6.3%	9.19	9.8%	9.67	8.8%
Best base-stock	6.73	0.8%	7.84	2.3%	8.60	2.9%	9.23	3.9%
Best constant				15.78				
(d) $p = 39$								
Optimal	7.84		9.11		10.04		10.79	
Myopic	7.88	0.6%	9.16	0.6%	10.17	1.3%	11.04	2.3%
Myopic-2	7.84	0.0%	9.12	0.1%	10.07	0.3%	10.84	0.4%
Dual-balancing	9.31	18.9%	10.73	17.8%	11.71	16.7%	12.61	16.9%
Std. vector base-stock	7.97	1.7%	9.29	2.0%	10.39	3.5%	11.38	5.4%
Better vector base-stock	7.84	0.1%	9.14	0.4%	10.14	1.0%	10.93	1.3%
Std. base-stock	7.98	1.8%	9.31	2.3%	10.43	3.9%	11.43	5.9%
Best base-stock	7.86	0.3%	9.19	0.9%	10.22	1.8%	11.06	2.5%
Best constant				18.21				

time. In many cases it is worse than the constant-order policy. The best one is certainly better, but it is still not very good. It is sometimes (Table 4(a) for $L = 3, 4$) worse than the constant-order policy. These policies perform better for shorter L and larger p .

The myopic policy is fairly good, and the myopic-2 policy is even better. They perform best for short lead times and large p . The dual-balancing policy is good for small p , but poor for large p . It actually improves as the lead time grows. All three methods perform better for lower variance (Poisson demand) than higher variance (geometric demand).

The standard vector base-stock policy provides a slight improvement over the standard (scalar) base-stock policy. Its performance is still poor, however. The better vector base-stock policy is indeed much better; it performs quite well in most instances. The case of Table 4(a) is an exception. Incidentally, results not reported here indicate that the best value of θ is almost always increasing in L . This observation suggests a heuristic: Start with small L , find the best θ , and use that as a starting point for larger L .

To summarize, even these limited tests indicate that base-stock policies do not perform well, except for short lead

Table 4. Average cost: Geometric demand.

Policy	Lead time							
	1		2		3		4	
(a) $p = 4$								
Optimal	9.82		10.24		10.47		10.61	
Myopic	9.95	1.4%	10.57	3.2%	10.99	5.0%	11.31	6.6%
Myopic-2	9.83	0.1%	10.29	0.5%	10.59	1.2%	10.79	1.7%
Dual-balancing	9.97	1.6%	10.34	1.0%	10.54	0.7%	10.67	0.6%
Std. vector base-stock	10.26	4.5%	11.72	14.4%	12.81	22.4%	13.86	30.6%
Better vector base-stock	9.87	0.6%	10.53	2.9%	11.26	7.6%	12.04	13.5%
Std. base-stock	10.49	6.8%	12.24	19.6%	13.55	29.5%	14.80	39.5%
Best base-stock	10.04	2.3%	10.70	4.5%	11.13	6.4%	11.44	7.8%
Best constant				11.00				
(b) $p = 9$								
Optimal	14.51		15.50		16.14		16.58	
Myopic	14.64	0.9%	15.93	2.8%	16.86	4.5%	17.61	6.2%
Myopic-2	14.51	0.0%	15.57	0.5%	16.32	1.1%	16.89	1.9%
Dual-balancing	15.61	7.6%	16.59	7.0%	17.11	6.0%	17.46	5.3%
Std. vector base-stock	14.98	3.3%	17.19	10.9%	18.96	17.5%	20.18	21.7%
Better vector base-stock	14.60	0.6%	15.66	1.0%	16.37	1.4%	16.83	1.5%
Std. base-stock	15.13	4.3%	17.53	13.1%	19.46	20.6%	20.78	25.3%
Best base-stock	14.73	1.5%	15.99	3.1%	16.87	4.6%	17.54	5.8%
Best constant				18.19				
(c) $p = 19$								
Optimal	19.22		20.89		22.06		22.95	
Myopic	19.37	0.8%	21.30	1.9%	22.79	3.3%	24.02	4.7%
Myopic-2	19.22	0.0%	20.95	0.3%	22.24	0.8%	23.28	1.4%
Dual-balancing	22.82	18.7%	24.60	17.8%	25.69	16.4%	26.46	15.3%
Std. vector base-stock	19.72	2.6%	22.31	6.8%	24.77	12.3%	26.68	16.2%
Better vector base-stock	19.31	0.5%	21.07	0.9%	22.35	1.3%	23.32	1.6%
Std. base-stock	19.81	3.1%	22.49	7.7%	25.06	13.6%	27.04	17.8%
Best base-stock	19.40	1.0%	21.31	2.0%	22.73	3.0%	23.85	3.9%
Best constant				28.60				
(d) $p = 39$								
Optimal	23.87		26.21		27.96		29.36	
Myopic	23.97	0.4%	26.55	1.3%	28.61	2.3%	30.31	3.3%
Myopic-2	23.87	0.0%	26.26	0.2%	28.10	0.5%	29.63	0.9%
Dual-balancing	31.72	32.9%	34.55	31.8%	36.67	31.2%	37.26	26.9%
Std. vector base-stock	24.43	2.4%	27.39	4.5%	30.06	7.5%	32.65	11.2%
Better vector base-stock	23.95	0.4%	26.40	0.7%	28.26	1.1%	29.77	1.4%
Std. base-stock	24.48	2.6%	27.48	4.8%	30.21	8.0%	32.82	11.8%
Best base-stock	24.00	0.6%	26.55	1.3%	28.51	2.0%	30.12	2.6%
Best constant				36.73				

times. One should give serious consideration to alternative methods. This echoes a point made by Hill (1999). Beyond this point, it is premature to draw firm conclusions about which approach is best.

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